Sparse Disapproval Guarantees a Nonempty Hare Core

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Abstract

An approval committee is Hare-core stable if no coalition meeting the Hare quota can strictly improve by moving to another candidate set. Whether every approval election has such a committee remains open. We prove nonemptiness when each voter disapproves at most two candidates, with no bounds on the numbers of candidates, seats, or voter types. The result also permits arbitrary positive rational voter weights. Our deterministic rule represents a committee by its missing set. It first maximizes weighted coverage of two-candidate disapproval sets and then maximizes total disapproval incidence. An exact coverage inequality excludes targets one seat below the committee. The incidence objective excludes unanimous equal-size targets, while targets of size at most $k-2$ cannot improve any voter. Two implementation-level independent verifiers audit overlapping finite grids. The symbolic proof, not this bounded enumeration, establishes the theorem's unbounded quantifiers. The argument identifies complement-side coverage as a tractable mechanism for a broad parameter range within a sharply defined preference domain.

1 Introduction

An approval election consists of a finite voter set $\mathcal{N}$, a candidate set C of size m, approval sets $A_i \subseteq C$, and a target committee size k. A nonempty set $T \subseteq C$ blocks a size-k committee W at the Hare quota when

$$\sum_{i:\,|A_i\cap T|>|A_i\cap W|} w_i \geq \frac{|T|}{k}, \tag{1}$$

where the positive rational voter weights sum to one. A committee is in the Hare core if it has no blocker. Targets may overlap the committee. For example, when $k=3$, a singleton target reaches its Hare quota at one third of the total voter weight. If all those voters strictly gain, it blocks; equality at the quota already permits blocking.

The core was introduced as a strong proportional-representation requirement for approval committees by Aziz et al. [1]. General deterministic nonemptiness is open. The publicly available results verified for this comparison establish nonemptiness for $k \leq 8$ or $m \leq 15$ [2], and for at most seven weighted voter types [3]. Other positive results impose interval or related preference structure [4]. Approval-based apportionment guarantees core stability on a multi-copy party domain [5]. That model constrains how candidates are grouped, rather than how many candidates each voter disapproves. Berker et al. [6] give an exact mixed-integer and duality framework with additional special cases.

Our restriction is different from these parameter and structural bounds. Write

$$H_i = C \setminus A_i$$

for voter i's disapproval set, and assume only $|H_i| \le 2$. This is a dense-approval or near-consensus regime, but our motivation is structural rather than empirical. Each disapproval set is an empty set, a vertex, or an edge. Selecting a missing candidate set therefore becomes a weighted vertex-edge coverage problem. This complement-side representation permits arbitrary m, k, and numbers of voter types. A targeted primary-source audit used "bounded disapproval," "two-veto," "dense approvals," and "complement rank" terminology. It found no equivalent theorem, so we describe the result only as apparently new after a targeted audit.

2 Selection rule and main theorem

The endpoint $m = k$ is immediate by taking $W = C$. Assume henceforth that $m > k \ge 2$ and put $s = m - k \ge 1$. A committee is represented by its missing set $B = C \setminus W$, where $|B| = s$. Define

$$e(B) = \sum_{i:\,|H_i|=2} w_i \, \mathbf{1}[H_i \cap B \neq \varnothing], \qquad j(B) = \sum_i w_i |H_i \cap B|. \tag{2}$$

Fix a total order on C. Choose a size-s set B lexicographically maximizing $(e(B), j(B))$. If several sets remain, choose the first under the induced lexicographic order on size-s subsets, and return $W = C \setminus B$.

Theorem 1 (Co-rank-two core theorem)**.** *For every $m \ge k \ge 2$, every approval election in which $|C \setminus A_i| \le 2$ for every positive-weight voter has a nonempty Hare core. When $m > k$, every committee associated with an (e, j)-maximizing missing set is in the core; in particular, the fixed tie-break returns a core committee.*

The rule is single-valued, explicit, and finite once the candidate order is fixed. We do not claim a polynomial-time implementation; exhaustive evaluation over all $\binom{m}{s}$ missing sets establishes the mathematical selection rule and is the method used by the supplied verifier.

3 Proof

A target of size $t > k$ cannot block because its quota t/k exceeds the total voter weight one. We may therefore restrict attention to a nonempty target T of size $t \le k$. Write $T = C \setminus D$ and $d = k - t$. Then $|D| = s + d$. Direct cancellation gives, for every voter,

$$|A_i \cap T| - |A_i \cap W| = -d + |H_i \cap D| - |H_i \cap B|. \tag{3}$$

3.1 Targets of size at most $k - 2$

If $d \ge 2$, the right side of (3) is at most $-d + |H_i| \le 0$. No voter strictly gains, so such a target cannot block.

3.2 Targets of size $k-1$

Now $d=1$. Strict gain in (3) requires

$$|H_i \cap D| - |H_i \cap B| \geq 2.$$

Thus every gainer has $|H_i|=2$, has $H_i \subseteq D$, and has $H_i \cap B = \varnothing$. In particular, every gainer is a two-disapproval voter not counted by $e(B)$.

Lemma 2. *A uniformly random size-s set meets any fixed two-set with probability*

$$p_{m,s} = 1 - \frac{k(k-1)}{m(m-1)} > \frac{1}{k}. \tag{4}$$

Proof. The avoidance probability is $\binom{m-2}{s}/\binom{m}{s} = k(k-1)/(m(m-1))$. With $m = k+s$,

$$k[m(m-1) - k(k-1)] - m(m-1) = (k-1)((2s-1)k + s(s-1)) > 0,$$

which is exactly the strict inequality in (4). □

Let w_2 be the total weight of two-disapproval voters. Averaging (2) over random B' and using maximality gives $e(B) \geq p_{m,s} w_2$. Hence every size-$(k-1)$ target has gainer weight at most

$$w_2 - e(B) \leq (1 - p_{m,s}) w_2 < 1 - \frac{1}{k} = \frac{k-1}{k}.$$

This is strictly below its Hare quota.

3.3 Targets of size k

Here $d=0$, so blocking requires every positive-weight voter to gain. Suppose that happened for some other size-s missing set D. Equation (3) yields

$$|H_i \cap D| > |H_i \cap B| \qquad \text{for every } i. \tag{5}$$

An empty H_i makes (5) impossible. For each two-disapproval voter, (5) implies that every pair hit by B is also hit by D, and every pair missed by B is hit by D. Therefore $e(D) \geq e(B)$. Strict inequality contradicts the first lexicographic objective. If equality holds, multiply (5) by the positive weight w_i and sum over all voters. This gives $j(D) > j(B)$, contradicting the tie-break. Thus no size-k target blocks. Together with the two preceding cases this proves Theorem 1.

4 Exact verification

The primary set-based verifier represents each anonymous profile as a tuple of immutable disapproval sets. For $m = 4, 5, 6$, it tests every $2 \leq k < m$ and every profile with total integer multiplicity at most four. For $m = 7$, the multiplicity bound is three. This grid contains 14 (m, k) rows and checks 101,851 profile–parameter cases, 236,801 tied maximizers, and 12,020,946 literal committee–target inequalities.

The implementation-level independent verifier uses bitmasks, weighted type-count vectors, and a separate enumeration architecture. It repeats the same 14 rows. It also adds $(m, k) = (3, 2)$ with multiplicity at most four and all $2 \leq k < 8$ at $m = 8$ with multiplicity at most two. It checks

106,620 profile–parameter cases, 265,371 tied maximizers, and 15,499,790 literal inequalities. Both implementations construct every tied maximizer, not merely one.

All verifier arithmetic is integral, and both scripts refuse to run when Python assertions are disabled. The grids enumerate only the stated small integer multiplicities. They do not exhaust arbitrary denominators of rational weights. Lemma 2 and the symbolic proof establish those unbounded quantifiers; the overlapping grids provide regression and falsification evidence only.

5 Scope and open boundary

The theorem permits arbitrary positive rational weights, repeated ballots, unapproved candidates, and targets overlapping the committee. Equality in the Hare quota counts as blocking throughout. The theorem does not cover a voter disapproving three candidates, does not resolve co-rank-three, and does not imply unrestricted core nonemptiness. Its contribution is a specialist restricted-domain theorem, not evidence that dense approvals are empirically universal. The complement-side coverage and tie-break mechanism explains why co-rank-two is tractable. Whether a higher-order objective can control deficit-one targets without losing the equal-size tie-break remains open. The present averaging proof alone does not supply such an extension.

Reproducibility and AI assistance

The package README gives a one-command exact replay and PDF build. GPT-5.6 Sol was used during proof exploration, verifier development, literature-query formulation, and manuscript preparation. The author reviewed the statements, proofs, citations, and executable artifacts and takes responsibility for the content.

Code availability

The verification code and reproducibility materials are available from the `https://github.com/Baymax-ray/approval-core-corank-two`.

References

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